Yuxuan Xia

Contact Information

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Research Interests and Statistics

My research interests lie broadly in Bayesian inference and machine learning, with a particular focus on sensor fusion and object tracking. My current work centers on multi-object tracking, multi-sensor fusion, and simultaneous localization and mapping using lidar and radar for autonomous systems. I am enthusiastic about exploring hybrid approaches to environmental perception that leverage deep learning-based and model-based methods.

Google Scholar stats (up to 2026 October): Citations: 2100+, h-index: 24, i10-index: 36.

Education

- **Ph.D. in Signals and Systems**
  Chalmers University of Technology, Gothenburg, Sweden, 2022
  Main supervisor: Prof. Lennart Svensson (RIP); Co-supervisor: Dr. Karl Granström.
  Doctoral thesis: *Poisson multi-Bernoulli mixtures for multiple object tracking.*

- **Lic. Eng. in Signals and Systems**
  Chalmers University of Technology, Gothenburg, Sweden, 2020
  Licentiate thesis: *Conjugate priors for Bayesian object tracking.*

- **M.Sc. in Communication Engineering**
  Chalmers University of Technology, Gothenburg, Sweden, 2017
  Supervisor: Dr. Karl Granström.
  Master thesis: *A study of Poisson multi-Bernoulli mixture conjugate prior in multiple target estimation.*

- **B.Sc. in Internet of Things Engineering**
  Jiangnan University, Wuxi, China, 2015

Working Experience

- **Research Assistant Professor**
  Smart Sensor Fusion Laboratory, School of Automation and Intelligent Sensing,
  Shanghai Jiao Tong University, Shanghai, China, 2024 Oct–

- **Industrial Postdoctoral Researcher**
  Zenseact (Volvo Cars) and Linköping University, Linköping, Sweden, 2023 Nov–2024 Sep
  Academic advisor: Prof. Gustaf Hendeby; Industrial advisor: Dr. Erik Stenborg.
  Project: The map to everywhere: Building HD maps for autonomous driving using crowdsourcing.

- **Postdoctoral Researcher**
  Chalmers University of Technology, Gothenburg, Sweden, 2022 Sep–2023 Oct
  Advisor: Prof. Lennart Svensson.

- **Research Intern**
  Mitsubishi Electric Research Laboratories, Cambridge, Massachusetts, U.S., 2019 Jun–2019 Dec
  Advisor: Dr. Pu (Perry) Wang, Dr. Karl Berntorp.
  Project: Extended object tracking with automotive radar.

- **Project Assistant**
  Chalmers University of Technology, Gothenburg, Sweden, 2017 Oct–2018 Jan

- **Software Engineer Intern**
  Remarkable Universal Network, Wuxi, China, 2014 Sep–2015 Mar

Teaching Experience

- **Information Fusion**, Graduate course (in English) at Shanghai Jiao Tong University, 2025–2026
  Role: Examiner, lecturer, course material development.

- **Multi-Object Tracking for Automotive Systems**, Graduate course on edX, 2018–2023
  Role: Coordinator, course material development.

- **Sensor Fusion and Nonlinear Filtering**, Graduate course at Chalmers, 2018–2021
  Role: Teaching Assistant, course material development.

- **Deep Machine Learning**, Graduate course at Chalmers, 2020
  Role: Teaching Assistant.

Supervised Theses

Master Theses

- *Targetless real-time spatial consistency estimation and optimization for heterogeneous information fusion*, Shanghai Jiao Tong University, 2026

- *Multi-object tracking based on multi-sensor multi-Bernoulli densities fusion*, Volvo Group, 2023

- *Model-based tracking of multiple merging and diverging highway lanes*, Zenseact, 2023

- *Deep decentralised multi-object tracking*, Chalmers, 2022

- *Deep extended object models*, Chalmers, 2022

- *Sequentially connected 2D assignments using Lagrangian dual methods*, Chalmers, 2022

- *Transforming the field of multi-object tracking*, Chalmers, 2021

Bachelor Theses

- *A systematic comparison study of deep learning based video instance segmentation techniques*, Shanghai Jiao Tong University, 2025

Presentation Experience

- **Invited Lecture**
  Data Association
  International Conference on Information Fusion (Bootcamp), 2025.

- **Invited Presentation**
  Multi-Target Tracking Using Sets of Trajectories
  IET International Radar Conference, 2025.

- **Tutorial**
  Poisson Multi-Bernoulli Mixtures for Multiple Target Tracking
  International Conference on Information Fusion 2020–2024.
  International Conference on Multisensor Fusion and Integration for Intelligent Systems 2024.
  Poisson Multi-Bernoulli Mixtures for Multiple Target Tracking and SLAM
  International Conference on Information Fusion 2025–2026.
  International Conference on Multisensor Fusion and Integration for Intelligent Systems 2025–2026.

- **Seminar**
  Data Association in Multiple Object Tracking: Challenges and Solutions
  KTH Royal Institute of Technology, Stockholm, 2025.
  Poisson Multi-Bernoulli Mixtures for Multi-Object Tracking
  Southeast University, Nanjing, 2024.
  University of Electronic Science and Technology of China, Sichuan, 2025.
  Bayesian Multi-Lane Tracking and Fusion Using Lane Marking Detections
  Zenseact, Gothenburg, 2024.
  University of Electronic Science and Technology of China, Sichuan, 2025.
  Conjugate Priors for Bayesian Object Tracking
  Zenseact, Gothenburg, 2021 and Linköping University, Linköping, 2022.

- **Paper Presentation**
  International Conference on Information Fusion 2017–2020, 2023–2024, 2026
  International Conference on Multisensor Fusion and Integration for Intelligent Systems 2024–2026.
  International Conference on Acoustics, Speech and Signal Processing 2020, 2023.
  Radar Conference 2020.
  Sensor Array and Multichannel Signal Processing Workshop 2020.

Grants and Awards

- **National Natural Science Foundation of China (Young Scientist Fund)**
  PI, 300,000 RMB, 2026–2028
  A prestigious national-level research grant supporting early-career researchers under the age of 35 to conduct independent and innovative research. It emphasizes scientific originality and the potential for academic advancement.
  Project: Simultaneous localization and mapping with 4D mmWave radar.

- **SJTU-Warwick Joint Seed Fund**
  PI, 50,000 RMB, 2026–2027
  A fund aims to foster new initiatives with high potential for global impact and towards novel research outcomes between Shanghai Jiao Tong University and University of Warwick (10 projects per year).
  Co-PI: Prof. Matthew D. Higgins at WMG, University of Warwick.
  Project: GNSS-anchored multipath SLAM and multi-object tracking with 5G signals.

- **Mengminwei International Exchange Fund**, 2026
  A dedicated fund of RMB 15,000 to support faculty members under the age of 45 at Shanghai Jiao Tong University in attending international conferences (10 awardees per year).

- **Shanghai Magnolia Youth Talent Program**
  PI, 750,000 RMB, 2025–2027
  A prestigious initiative by the Shanghai municipal government to support outstanding young professionals and scholars contributing to the city's innovation and development.
- **Industrial postdoctoral fellowship**
  Wallenberg AI Autonomous System and Software Program, PI, 2,200,000 SEK, 2023
  Project: The map to everywhere: Building HD maps for autonomous driving using crowdsourcing.
- **China International College Students' Innovation Competition**
  Supervised project: Smart sensor fusion data center for autonomous systems. Silver medal.
- **Paper awards**
  International Conference on Multisensor Fusion and Integration for Intelligent Systems, 2026, Two best student paper awards (first runner-up, second runner-up)
  International Conference on Multisensor Fusion and Integration for Intelligent Systems, 2024, Best paper award (first runner-up)
  International Conference on Information Fusion, 2021, Best student paper award
  Sensor Array and Multichannel Signal Processing Workshop, 2020, Best paper award finalist
- **International travel award**
  International Conference on Information Fusion, 2017
- **Adlerbert 60th anniversary scholarships**
  Chalmers University of Technology, 60000 SEK, 2016
  Awarded based on high GPA (17 awardees in total).
- **Avancez scholarships**
  Chalmers University of Technology, 85% tuition fee reduction, 2015

Professional Service

- **Conference Session Chair**
  FUSION 2023, MFI 2026
- **Conference Reviewer**
  FUSION; IEEE MFI, ICASSP, IVS, ICITS, ICRA
- **Journal Reviewer**
  IEEE T-SP, T-AES, T-ITS, T-IV, T-VT, T-ASE, T-WC, T-MC, T-IP, T-NNLS, T-NSE, T-CDS, T-PAMI. IEEE/CAA J. Automatica Sinica, IEEE OJSP, Sensors Journal, Access, RAL, SPL, SL
  Elsevier IF, EAAI, PR, KBS, SP, DSP, IS; IET SP, RSN; ISIF JAIF

Languages and Coding Skills

- **Languages**: Chinese (mother tongue), English (Fluent in speech and writing).
- **Coding Skills**: familiar with MATLAB, Python (PyTorch); experiences in C/C++, ROS, C#, SQL.

Journal Publications

1. *__Yuxuan Xia__, Ángel F. García-Fernández, Lennart Svensson.
   Efficient implementations of extended object PMBM filters with blocked Gibbs sampling.
   IEEE Transactions on Aerospace and Electronics Systems, 62, 18260–18277, 2026.
2. Ángel F. García-Fernández, Jinhao Gu, Lennart Svensson, **Yuxuan Xia**, Jan Krejčí, Oliver Kost, Ondřej Straka.
   GOSPA and T-GOSPA quasi-metrics for evaluation of multi-object tracking algorithms.
   IEEE Transactions on Aerospace and Electronics Systems, 62, 9631–9644, 2026.
3. Jan Krejčí, Oliver Kost, **Yuxuan Xia**, Lennart Svensson, Ondřej Straka.
   Occlusion-aware multi-object tracking via expected probability of detection.
   IEEE Transactions on Aerospace and Electronics Systems, 62, 8211–8228, 2026.
4. Tao Liang, Lin Gao, Liangliang Li, Chaoqun Yang, Ting Yuan, **Yuxuan Xia**, Ping Wei.
   Tensor decomposition enhanced LRFS multi-DOA tracking based on coprime array.
   IEEE Transactions on Aerospace and Electronics Systems, 62, 4586–4600, 2026.
5. Lin Chen, Lin Gao, **Yuxuan Xia**, Chaoqun Yang, Zijie Shang, Zhicheng Su, Ting Yuan, Ping Wei.
   Possibility PMBM filter for robust multi-target tracking.
   Signal Processing, 110483, 2026.
6. Guanhua Ding, **Yuxuan Xia**, Runwei Guan, Qinchen Wu, Tao Huang, Weiping Ding, Jinping Sun, Guoqiang Mao.
   OptiPMB: Enhancing 3D multi-object tracking with optimized Poisson multi-Bernoulli filtering.
   IEEE Transactions on Intelligent Transportation Systems, 26(12), 22312–22328, 2025.
7. Tao Huang, Jianan Liu, Xi Zhou, Dinh C. Nguyen, Mostafa Rahimi Azghadi, **Yuxuan Xia**, Qing-Long Han, Sumei Sun.
   Vehicle-to-everything cooperative perception for autonomous driving.
   Proceedings of the IEEE, 113(5), 443–477, 2025.
8. Zhenrong Zhang, Jianan Liu, **Yuxuan Xia**, Tao Huang, Qing-Long Han, Hongbin Liu.
   LEGO: Learning and graph-optimized modular tracker for online multi-object tracking with point clouds.
   IEEE Transactions on Circuits and Systems for Video Technology, 36(2), 2419-2432, 2025.
9. *__Yuxuan Xia__, Ángel F. García-Fernández, Johan Karlsson, Kuo-Chu Chang, Ting Yuan, Lennart Svensson.
   Probabilistic GOSPA: A metric for performance evaluation of multi-object filters with uncertainties.
   IEEE Transactions on Aerospace and Electronics Systems, 61(5), 15113–15121, 2025.
10. Jiayin Deng, Zhiqun Hu, **Yuxuan Xia**, Zhaoming Lu, Xiangming Wen.
    Roadside 3-D extended object tracking by fusing sparse radar points and pixel keypoints.
    IEEE Sensors Journal, 13(1), 26132–26151, 2025.
11. Yu Ge, Ossi Kaltiokallio, **Yuxuan Xia**, Ángel F. García-Fernández, Hyowon Kim, Jukka Talvitie, Mikko Valkama, Henk Wymeersch, Lennart Svensson.
    Batch SLAM with PMBM data association sampling and graph-based optimization.
    IEEE Transactions on Signal Processing, 73, 2139–2153, 2025.
12. Karl Granström, Lennart Svensson, **Yuxuan Xia**, Jason L. Williams, Ángel F. García-Fernández.
    Poisson multi-Bernoulli mixtures for sets of trajectories.
    IEEE Transactions on Aerospace and Electronics Systems, 61(2), 5178–5194, 2025.
13. *__Yuxuan Xia__, Ángel F. García-Fernández, Lennart Svensson.
    Markov chain Monte Carlo data association for sets of trajectories.
    IEEE Transactions on Aerospace and Electronics Systems, 60(6), 7804–7819, 2024.
14. Hyowon Kim, Ángel F. García-Fernández, Yu Ge, **Yuxuan Xia**, Lennart Svensson, Henk Wymeersch.
    Set-type belief propagation with applications to Poisson multi-Bernoulli SLAM.
    IEEE Transactions on Signal Processing, 72, 1989–2005, 2024.

15. Weiyi Xiong, Jianan Liu, Tao Huang, Qing-Long Han, **Yuxuan Xia**, Bing Zhu.
LXL: LiDAR excluded lean 3D object detection with 4D imaging radar and camera fusion.
IEEE Transactions on Intelligent Vehicles, 9(1), 79-92, 2024.

16. *__Yuxuan Xia__, Ángel F. García-Fernández, Florian Meyer, Jason L. Williams, Karl Granström, Lennart Svensson.
Trajectory PMB filters for extended object tracking using belief propagation.
IEEE Transactions on Aerospace and Electronic Systems, 59(6), 9312–9331, 2023.

17. Juliano Pinto, Georg Hess, William Ljungbergh, **Yuxuan Xia**, Henk Wymeersch, Lennart Svensson.
Deep learning for model-based multi-object tracking.
IEEE Transactions on Aerospace and Electronic Systems, 59(6), 7363–7379, 2023.

18. Ángel F. García-Fernández, **Yuxuan Xia**, Lennart Svensson.
Poisson multi-Bernoulli mixture filter with general target-generated measurements and arbitrary clutter.
IEEE Transactions on Signal Processing, 71, 1895–1906, 2023.

19. Jianan Liu, Liping Bai, **Yuxuan Xia**, Tao Huang, Bing Zhu, Qing-Long Han.
GNN-PMB: A simple but effective online 3D multi-object tracker without bells and whistles.
IEEE Transactions on Intelligent Vehicles, 8(2), 1176–1189, 2022.

20. *__Yuxuan Xia__, Lennart Svensson, Ángel F. García-Fernández, Jason L. Williams, Daniel Svensson, Karl Granström.
Multiple object trajectory estimation using backward simulation.
IEEE Transactions on Signal Processing, *70*, 3249–3263, 2022.

21. Jianan Liu, Weiyi Xiong, Liping Bai, **Yuxuan Xia**, Tao Huang, Wanli Ouyang.
Deep instance segmentation with automotive radar detection points.
IEEE Transactions on Intelligent Vehicles, 8(1), 84–94, 2022.

22. *__Yuxuan Xia__, Karl Granström, Lennart Svensson, Maryam Fatemi, Ángel F. García-Fernández, Jason L. Williams.
Poisson multi-Bernoulli approximations for multiple extended object filtering.
IEEE Transactions on Aerospace and Electronic Systems, *58*(2), 890–906, 2022.

23. Juliano Pinto, **Yuxuan Xia**, Lennart Svensson, Henk Wymeersch.
An uncertainty-aware performance measure for multi-object tracking.
IEEE Signal Processing Letters, 28, 1689–1693, 2021.

24. Ángel F. García-Fernández, Jason L. Williams, Lennart Svensson, **Yuxuan Xia**.
A Poisson multi-Bernoulli mixture filter for coexisting point and extended targets.
IEEE Transactions on Signal Processing, 69, 2600–2610, 2021.

25. Ángel F. García-Fernández, Lennart Svensson, Jason L. Williams, **Yuxuan Xia**, Karl Granström.
Trajectory Poisson multi-Bernoulli filters.
IEEE Transactions on Signal Processing, 68, 4933–4945, 2021.

26. *__Yuxuan Xia__, Pu Wang, Karl Oskar Erik Berntorp, Lennart Svensson, Karl Granström, Hassan Mansour, Petros T. Boufounos, Philip Orlik.
Learning-based extended object tracking using hierarchical truncation measurement model with automotive radar.
IEEE Journal of Selected Topics in Signal Processing, *15*(4), 1013–1029, 2021.

27. *__Yuxuan Xia__, Karl Granström, Lennart Svensson, Ángel F. García-Fernández, Jason L. Williams.
Multi-scan implementation of the trajectory Poisson multi-Bernoulli mixture filter.
Journal of Advances in Information Fusion, *14*(2), 213–235, 2019.

28. Karl Granström, Lennart Svensson, Stephan Reuter, **Yuxuan Xia**, Maryam Fatemi.
Likelihood-based data association for extended object tracking using sampling methods.
IEEE Transactions on Intelligent Vehicles, 3(1), 30–45, 2017.

Conference Publications

1. Guofei Chen, Xiaojun Chen, **Yuxuan Xia**, Longlong Lin, Ting Yuan.
   VAET: Visibility-aware extent transformer for reliability-aware heterogeneous sensor fusion.
   Symposium Sensor Data Fusion (SDF): Trends, Solutions and Applications, 2026.
2. Gyuwoong Han, **Yuxuan Xia**, Yu Ge, Hyowon Kim.
   RCS Estimation and Exploitation for Monostatic OFDM Sensing.
   IEEE Annual Integrated Sensing and Communication Conference (ISAC), 2026.
3. Jan Krejčí, Oliver Kost, **Yuxuan Xia**, Lennart Svensson, Ondřej Straka.
   Poisson prior and principled occlusion handling.
   International Conference on Multisensor Fusion and Integration for Intelligent Systems, 2026.
4. Xueqi Qiu, ***Yuxuan Xia**, Hyowon Kim, Chaoqun Yang.
   An extended object PMB filter with zero-inflated Poisson model using belief propagation.
   International Conference on Multisensor Fusion and Integration for Intelligent Systems, 2026.
5. Chenyu Chen, Lin Gao, **Yuxuan Xia**, Chaoqun Yang, Wanchun Li, Ting Yuan, Ping Wei
   Superelliptical extended object tracking using variational Bayesian inference.
   International Conference on Information Fusion, 2026.
6. ***Yuxuan Xia**, Kuochu Chang, Xueqi Qiu, Lin Gao, Chaoqun Yang, Ting Yuan.
   Event-triggered distributed target tracking via PRIMEX.
   International Conference on Information Fusion, 2026.
7. Ángel F. García-Fernández, **Yuxuan Xia**.
   Variational PMB filter via coordinate descent Kullback-Leibler divergence minimisation.
   International Conference on Information Fusion, 2026.
8. Yiping Xie, ***Yuxuan Xia**, Erik Stenborg, Junsheng Fu, Axel Beauvisage, Gabriel E Garcia, Tianyu Wu, Gustaf Hendeby.
   B^2F-Map: Crowd-sourced mapping with Bayesian B-spline fusion.
   International Conference on Robotics and Automation, 2026.
9. ***Yuxuan Xia**, Ángel F. García-Fernández, Johan Karlsson, Yu Ge, Lennart Svensson, Ting Yuan.
   Probabilistic trajectory GOSPA: A metric for uncertainty-aware multi-object tracking performance evaluation.
   International Conference on Multisensor Fusion and Integration for Intelligent Systems, 2025.
10. Alfred Wärnsäter, **Yuxuan Xia**, Yuan Ting, Johan Karlsson.
    Efficient approximation of the trajectory GOSPA metric via graph-structured optimal transport.
    IEEE Radar Conference, 2025.
11. Ting Yuan, Xiaojun Chen, ***Yuxuan Xia**, Lifeng Zhang, Juergen Dickmann.
    Low-level data analysis for high-resolution automotive radar perception in real-world scenarios.
    IEEE International Radar Conference, 2025.
12. Jan Krejčí, Oliver Kost, **Yuxuan Xia**, Lennart Svensson, Ondřej Straka.
    Model-based multi-object visual tracking: Identification and standard model limitations.
    International Conference on Information Fusion, 2025.
13. Yu Ge, Ossi Kaltiokallio, Hui Chen, Jukka Talvitie, **Yuxuan Xia**, Giyyarpuram Madhusudan, Guillaume Larue, Lennart Svensson, Mikko Valkama and Henk Wymeersch.
    Target handover in distributed integrated sensing and communication.
    IEEE International Conference on Communications, 2025.
14. ***Yuxuan Xia**, Ángel F. García-Fernández, Lennart Svensson.
    Hybrid PHD-PMB trajectory smoothing using backward simulation.
    International Conference on Multisensor Fusion and Integration for Intelligent Systems, 2024.
15. ***Yuxuan Xia**, Erik Stenborg, Junsheng Fu, Gustaf Hendeby.
    Bayesian simultaneous localization and multi-lane tracking using onboard sensors and a SD map.
    International Conference on Information Fusion, 2024.

16. Guanhua Ding, Jianan Liu, **Yuxuan Xia**, Tao Huang, Bing Zhu, Jinping Sun.
LiDAR point cloud-based multiple vehicle tracking with probabilistic measurement-region association.
International Conference on Information Fusion, 2024.

17. Yuchuan Jin, Theodor Stenhammar, David Bejmer, Axel Beauvisage, **Yuxuan Xia**, Junsheng Fu.
Towards accurate ego-lane identification with early time series classification.
International Conference on Information Fusion, 2024.

18. Jianan Liu, Guanhua Ding, **Yuxuan Xia**, Jinping Sun, Tao Huang, Lihua Xie, Bing Zhu.
Which framework is suitable for online 3D multi-object tracking for autonomous driving with automotive 4D imaging radar?
IEEE Intelligent Vehicles Symposium, 2024.

19. *__Yuxuan Xia__, Ángel F. García-Fernández, Lennart Svensson.
An efficient implementation of the extended object trajectory PMB filter using blocked Gibbs sampling.
International Conference on Information Fusion, 2023.

20. Lechi Li, Chen Dai, *__Yuxuan Xia__, Lennart Svensson.
Deep fusion of multi-object densities using transformer.
IEEE International Conference on Acoustics, Speech and Signal Processing, 2023.

21. Weiyi Xiong, Jianan Liu, **Yuxuan Xia**, Tao Huang, Bing Zhu, Wei Xiang.
Contrastive learning for automotive mmWave radar detection points based instance segmentation.
IEEE International Conference on Intelligent Transportation Systems, 2022.

22. Ángel F. García-Fernández, **Yuxuan Xia**, Lennart Svensson.
A comparison between PMBM Bayesian track initiation and labelled RFS adaptive birth.
International Conference on Information Fusion, 2022.

23. Juliano Pinto, Georg Hess, William Ljungbergh, **Yuxuan Xia**, Lennart Svensson, Henk Wymeersch.
Next generation multitarget trackers: random finite set methods vs transformer-based deep learning.
International Conference on Information Fusion, 2021.

24. *__Yuxuan Xia__, Pu Wang, Karl Berntorp, Petros T. Boufounos, Philip V. Orlik, Lennart Svensson, Karl Granström.
Extended object tracking with automotive radar using learned structural measurement model.
IEEE Radar Conference, 2020.

25. *__Yuxuan Xia__, Lennart Svensson, Ángel F. García-Fernández, Karl Granström, Jason L. Williams.
Backward simulation for sets of trajectories.
International Conference on Information Fusion, 2020.

26. Ángel F. García-Fernández, Lennart Svensson, Jason L. Williams, **Yuxuan Xia**, Karl Granström.
Trajectory multi-Bernoulli filters for multi-target tracking based on sets of trajectories.
International Conference on Information Fusion, 2020.

27. *__Yuxuan Xia__, Pu Wang, Karl Berntorp, Hassan Mansour, Petros T. Boufounos, Philip V. Orlik.
Extended object tracking using hierarchical truncation model with partial-view measurements.
IEEE Sensor Array and Multichannel Signal Processing Workshop, 2020.

28. *__Yuxuan Xia__, Pu Wang, Karl Berntorp, Toshiaki Koike-Akino, Hassan Mansour, Milutin Pajovic, Petros T. Boufounos, Philip V. Orlik.
Extended object tracking using hierarchical truncation measurement model with automotive radar.
IEEE International Conference on Acoustics, Speech and Signal Processing, 2020.

29. Karl Granström, Lennart Svensson, **Yuxuan Xia**, Ángel F. García-Fernández, Jason L. Williams.
Spatiotemporal constraints for sets of trajectories with applications to PMBM densities.
International Conference on Information Fusion, 2020.

30. *__Yuxuan Xia__, Karl Granström, Lennart Svensson, Ángel F. García-Fernández, Jason L. Williams.
Extended target Poisson multi-Bernoulli mixture trackers based on sets of trajectories.
International Conference on Information Fusion, 2019.

31. Ángel F. García-Fernández, **Yuxuan Xia**, Karl Granström, Lennart Svensson, Jason L. Williams.
Gaussian implementation of the multi-Bernoulli mixture filter.
International Conference on Information Fusion, 2019.

32. Karl Granström, Lennart Svensson, **Yuxuan Xia**, Jason L. Williams, Ángel F. García-Fernández.
Poisson multi-Bernoulli mixture trackers: Continuity through random finite sets of trajectories.
International Conference on Information Fusion, 2018.

33. *__Yuxuan Xia__, Karl Granström, Lennart Svensson, Ángel F. García-Fernández.
An implementation of the Poisson multi-Bernoulli mixture trajectory filter via dual decomposition.
International Conference on Information Fusion, 2018.

34. *__Yuxuan Xia__, Karl Granström, Lennart Svensson, Ángel F. García-Fernández.
Performance evaluation of multi-Bernoulli conjugate priors for multi-target filtering.
International Conference on Information Fusion, 2017.

Patents

1. *System and method for tracking expanded state of an object.*
US Patent 11,619,494.

2. *System and method for tracking expanded state of moving object with model geometry learning.*
US Patent 11,879,964.

3. *Model-based road estimation.*
US Patent 12,668,247.

4. *Bayesian lane-level map fusion using B-splines.*
European Patent App. 25202187.8